

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

MACHINE LEARNING [IT-4402]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify three common issues found in raw datasets and suggest appropriate methods to clean them. (5)
 - A)
 - B) Describe the role of feature scaling in data preprocessing and compare normalization and standardization. (5)
- 2) Suppose you are working on an email spam detection system. Explain how supervised learning and logistic regression can be used to classify emails as spam or not spam. (5)
 - A)
 - B) Consider a scenario where a bank wants to predict loan defaults. How can boosting techniques like AdaBoost or Gradient Boosting help improve prediction accuracy? (5)
- 3) Explain the bias-variance tradeoff with the help of an example. Why is it important to find the right balance? (5)
 - A)
 - B) Explain how the VC dimension of a hypothesis class helps in understanding the capacity of a learning model. Provide an example. (5)
- 4) Explain the significance of clustering in real-world applications. Provide two examples where clustering is used. (5)
 - A)
 - B) Explain how ICA is used in blind source separation problems, such as separating mixed audio signals. (5)
- 5) Explain the Bellman equation, and how important is it in reinforcement learning? (5)
 - A)
 - B) Given a system with uncertainty and noise, how would you apply Linear Quadratic Gaussian (LQG) control to improve decision-making? (5)

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